IIT ROORKEE

FREE ONLINE EDUCATION swayam शिक्षित भारत, उन्नत भारत

NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 19: EIA Case study - III

By-
Prof. Bhola Ram Gurjar
Dept. of Civil Engineering

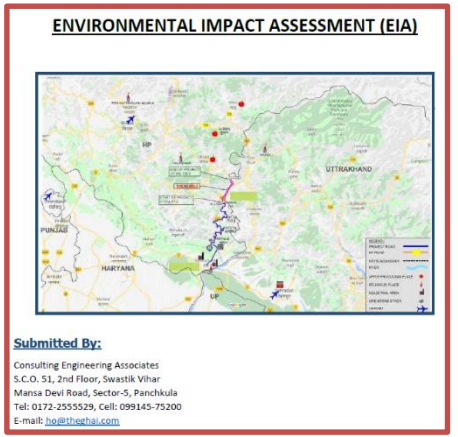


Contents

- Case study for a hilly road {NH-707, Himachal Pradesh}
  - Project details
  - Benefits
  - Steps of EIA process
  - Features of project area
  - Environment policies & regulations, clearance for project
  - Public participation & redressal of issues
  - Impacts & mitigation measures

EIA report of hilly road in the state of Himachal Pradesh

- Gumma to Fediz (NH-707)
- The Ministry of Road Transport and Highways to improve interconnectivity has undertaken development of many single or intermediate lane National Highways.
- Under National Highways Interconnectivity Project (NHIIP) aided by the World Bank
- EIA report is prepared by ‘Consulting Engineers Associates, Panchkula’

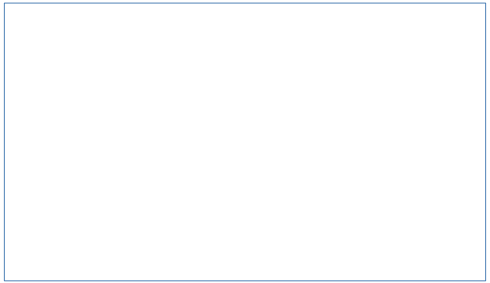

ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

Submitted By:

Consulting Engineering Associates
S.C.O. 51, 2nd Floor, Swastik Vihar
Mansa Devi Road, Sector-5, Panchkula
Tel: 0172-2555529, Cell: 099145-75200
E-mail: hq@theghai.com

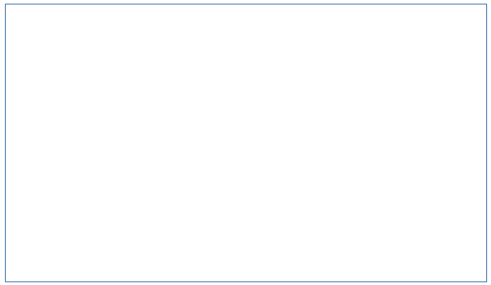
Benefits of the project

- Reduction of Vehicle Operating Cost (VOC) by reducing road roughness and improved alignment.
- Improving financial sustainability for road development and maintenance.
- Increased vehicle speeds.
- Reduced passenger time costs.



Steps for EIA report

- Study of Background information on project
- Screening report Preparation
- Survey Part-1 (Baseline Survey or Baseline Data Collection)
- Survey Part-2 (socio-economic survey)
- Analysis of practical Impacts
- Meeting with concerned Stakeholders
- Mitigation Measures Adoption
- Environment Management Plan



Physiographical features of project area

- Most of the areas form a part of hilly tract with undulating terrain
- Elevation varies from 2500m to 3500m above mean sea level
- The project influence area is marked by deep valleys, river terraces, steep escapements and innumerable streams



Other physical features of project ar



Geological:

- Primarily consists of alluvial deposits.
- Project area also comprise of sandstone.
- Primarily clay forms the top zone of area.
- Presence of variable sizes and types of rocks in the area

Seismology:

- Moderately earthquake prone.
- Earthquakes have been experienced in many parts of the state

Landslide in project area

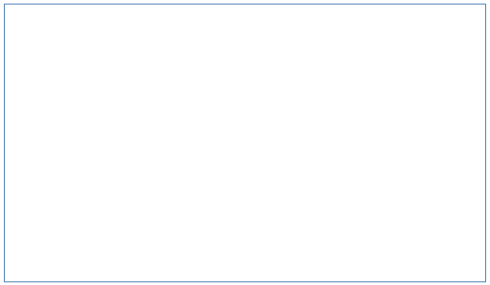
- There is major issue of soil stability in the hilly terrain of the project area.
- Accompanied by landslips, mud flows and flash floods affecting the entire area.
- The problem rises during rainy season and adversely affects the road structure.

S. No.	FROM (Km)	UPTO (Km)
1	95.530	95.580
2	95.400	95.430
3	96.520	96.570
4	98.800	98.850
5	101.170	101.270
6	101.370	101.470
7	101.690	101.730
8	102.630	102.690

Table: Major landslide identified points (in terms of Chainage)

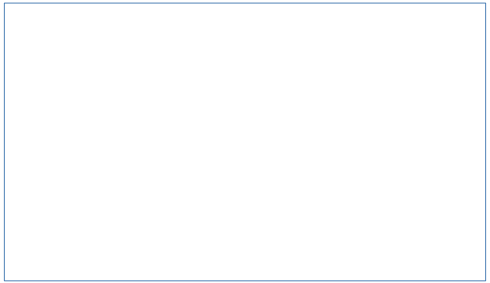
Environmental policies applicable for the project

- The National Forest Policy, 1988
- National Water Policy, 2012
- National Environment Policy, 2006
- National Conservation Strategy, 1992
- National Policy on Resettlement and Rehabilitation for Project Affected Families, 2007



Environmental laws & regulation applicable for project

- The Environment (Protection) Act, 1986
- The Forest (Conservation) Act, 1980 (amended in 2003 & 2004)
- The Wildlife (Protection) Act (amended in 2002)
- The Air (Prevention and Control of Pollution) Act, 1972
- The Water (Prevention & Control of Pollution) Act, 1974
- The EIA Notification, 2006 (& amendments)



Environmental clearance requirements 1/2

Government of India:

- This project is widening of existing State Highway
- Doesn't fall under any category for the environment clearance as per the EIA notification 2006

World Bank:

- Project has been classified as Category 'B' project
- Cumulative impacts from the project on the environmental and social components

Environmental clearance requirements 2/2

- State Ground Water Board:

Clearance required for extraction of ground water.

- State pollution control board:

Consent for the Establishment and Operation must be taken during the Construction stage of the Project for the operation of all Construction Plants.

- Forest Clearance:

From the Forest department for cutting of the trees

Comparison between ‘with project’ & ‘without project’ Scenario 1/2

Component	Without project	With project
• Connectivity & development	• Connectivity between the other National and State Highways is not good. • Pavement conditions are poor resulting in high travel time, bad vehicle conditions, and more pollution. • Poor pavement conditions restrict people’s access to local market, Medical facilities, education centers, work zones etc.	• Improve connectivity. • Travel time will be reduced • Pollution levels will be reduced as well. • Trade of local produce like dairy products, agriculture products, and small-scale industrial product will also be increased.

Comparison between ‘with project’ & ‘without project’ Scenario 2/2

Component	Without project	With project
• Flora	There will not be any impact on existing terrestrial ecology.	Felling of 186 trees proposed widening. Compensatory Afforestation will be done.
• Social & cultural environment	There will not be any impact on existing social and cultural environment.	03 structures (Encroachments) but none of the common property resources will be affected

Public Consultation

Project Corridor	Location	Chainage	No. of participants	Typical Issues
Gumma to Fediz Section of NH-707	Antroli Village	98+300	18	• Drainage problem • Road Safety • Bus Stop • Dust suppression measures should be adopted • Provision of speed breaker • Relocation of Structures • Junction Improvement
	Near Starting Point Gumma Village	95/000	16	



Sample sheet of public consultation documentation

PUBLIC/STAKEHOLDER'S CONSULTATION MEETING ATTENDANCE SHEET

Project	Gumma-Fediz section (NH 707)			Antpol P- Panchayat - Pawan.			
District		Taluka		Block/ward No		GP/MC	
Settlement		PS/Thana		Chainage/km		Venue	
Date	22-09-2019.	Time		Total No. of participants	Male	Female	Total

S. No	Name	Age	Gender	Contact Details	Occupation	Caste	Signature
1.	Virender Singh	32		9805267773			
2.	Anil Kumar	30		8894380899			
3.	KEWAL RAM	33		9805163066			
4.	Bhagmal	40		9805730780			
5.	Hema Singh	36		8894238891			Hema Singh
6.	Pawan Singh	35		9816719060			
7.	Prabhat Chauhan	41		9805887609			
8.							
9.							
10.							
11.							
12.							
13.							
14.							
15.							
16.							
17.							
18.							



Redressal of issues raised at public meeting 1/3

Issue	Redressal
• Loss of Income restoration Option	• People will be compensated adequately
• Road Safety	• Traffic calming measures (at all intersections), crossing points have been proposed for the safety of local people. • Hazard markers, RRPM (Raised Reflective Pavement Marker), road marking etc. have also been provided. • Crash barrier has also been proposed throughout the alignment



Hazard markers, RRPM (Raised Reflective Pavement Marker)



- Hazard markers are signs on road or adjacent to road showing upcoming bridge piers or traffic islands etc.
- These markers are precautionary traffic signs.

Back



- RRPM are used to maintain traffic direction for ongoing vehicle.
- Also dictates the limit of pavement (road)
- Made of reflective material to increase visibility during nighttime.

Crash Barriers





- Crash barrier are steel structure along the road.
- Stop or limit the vehicle from going off the road in case of accident/crash.
- Especially useful in hilly roads where it stops vehicle from going down to valley side in case of accident or imbalance.

Back

Redressal of issues raised at public meeting 2/3

Issue	Redressal
• Land Acquisition and Mode of Compensation	• Compensation will be made as per RAP*
• Flood Problem	• Total 38 nos. culverts have been provided at various locations
• Drainage	• Drainage has been provided throughout the alignment
• Loss of Trees	• Compensatory forestation would be done at the ratio of two trees for each tree to be cut. • Local species of trees have been selected for plantation



*RAP: Resettlement Action Plan (made as per regulations)

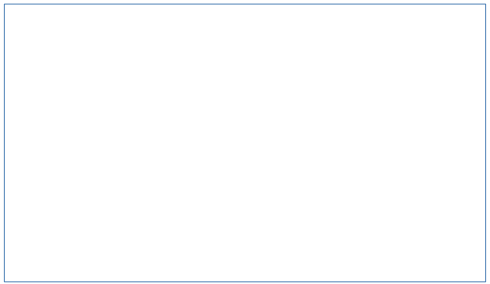
Redressal of issues raised at public meeting 3/3

Issue	Redressal
• Impact on Health	• Further study to assess the impact of bypass on community's health • Roadside plantation has been proposed to screen emissions from the traffic reaching the settlement areas
• Utilities and Basic Infrastructure	• All the utilities and basic infrastructures to be impacted will be relocated under the project cost.
• Employment during construction	• The locals will be given preference for employment during construction



Major impact of project on the environment

- Soil erosion & landslides
- Deforestation
- Flooding & drainage
- Air pollution
- Noise pollution
- Land use change
- Road safety



Soil erosion: Impact & mitigation measures

Impact	Mitigation (design)	Mitigation (implementation)
• Soil erosion	• Identification of erodible soils & Treatment of embankment slopes and stream inlet and outlet.	• Close control of timing of embankment treatment after earthwork operations.

Soil erosion and possibility of land slid is the biggest challenge & concern for construction project on hilly terrain

Rock Anchoring (Nailing) for slope stabilization: Mitigation measure

- Rock anchors or rock bolts is a long anchor bolt, for stabilizing rock excavations
- Application in tunnels or rock cuts
- Transfers load from unstable exteriors to the confined interior of the rock mass



Hydro-seeding for slope stabilization: Mitigation measure

Method of seeding that consists of applying mixture of :

- Water
- Seed
- wood fiber
- soil stabilizer (if used)

through hydroseeding equipment





Image source: finncorp.com

Bamboo net plantation for slope stabilization



- For dumping sites to check sliding of dumped material.
- Natural barrier for material (like soil or rock or waste)

Hedge brush layer for slope stabilization: Mitigation measure

- Plantation of local shrubs along the slope.
- In the landslide zone or places having soft rock/ soil
- To protect the hill slope from crumbling/weathering.
- Provided over the embankment on valley side (except the stretches in built-up area, Bridges and Culverts)



Deforestation: Impact & mitigation measures

Impact	Mitigation (design)	Mitigation (implementation)
• Tree cutting	• Minimize by alignment selection for proposed ROW (Right of Way).	• Control tree cutting where possible. • Afforestation programme in 1:3 ratio.

Flooding & drainage: Impact & mitigation measures

Impact	Mitigation (design)	Mitigation (implementation)
• Flooding/Drainage	• Total 38 nos. culverts provided at various locations. • Drainage has been provided throughout the alignment	• Keep channels clear prior to monsoon period.

Air pollution: Impact & mitigation measures

Impact	Mitigation (design)	Mitigation (implementation)
• Air pollution	• Wider road with paved shoulders will reduce use of unpaved shoulders. • Diversions and haul roads	• Setting of work areas/camp site away from sensitive receptors. • Modern well maintained construction equipment. • Dust suppression measures on plant like maximum paved area, sprinkling of water etc.

Noise pollution: Impact & mitigation measures

Impact	Mitigation (design)	Mitigation (implementation)
• Noise pollution	• Selected flexible pavement. • Vegetation barrier for noise reduction at source	• Setting of work areas away from sensitive receptors. • Use of well-maintained equipment and machinery. • Acoustic enclosures for DG sets.

Land use: Impact & mitigation measures

Impact	Mitigation (design)	Mitigation (implementation)
• Land use	• Shortest alignment for corridor where social disruption outweighs loss of agricultural land.	• Reinstate site working areas on completion of works.

Road safety: Impacts & mitigation measures

Impact	Mitigation (design)	Mitigation (implementation)
• Road Safety	• Median for all section to avoid head on collision • Traffic calming measures near junctions and village area • Pedestrian crossings • Crash barriers throughout the alignment • High standard of road signs, hazard marker and markings	• Provide safe traffic management methods

Monitoring of impacts on air & water quality

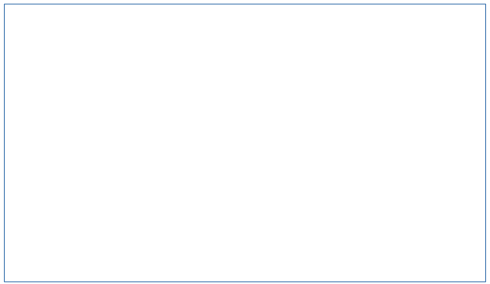
Impact	Parameters	Standards	Frequency
• Air	CO, NOx, SPM, RPM, and SO2	Air (prevention & Control of Pollution) Rules, CPCB, 1994	• Quarterly
• Water	• Physical, chemical and Biological parameters	• Indian Standards for Inland Surface Waters (IS: 2296, 1982) and for Drinking Water (IS : 10500 -1991)	• Quarterly

Monitoring of impacts on noise level, forestation & soil

Impact	Parameters	Standards	Frequency
• Noise	• noise level in decibel	MoEF&CC Noise Rules, 2000	• Quarterly
• Afforest-ation	• Survival rate	• -	• Quarterly
• Soil	• Pb (lead) • SAR (Sodium adbortion ration) • Oil & Grease	• Threshold for each contaminant set by IRIS database of USEPA	• Half-yearly

Conclusion

- Project is economically very beneficial for the local population with very little negative social impact.
- Soil erosion, possible landslides are the key concerns.
- Proper implementation of mitigation measures will ensure very little impact on environment.



References:

- Government of Himachal Pradesh, Public Works Department (PWD), NH-Division; “EIA Report- Rehabilitation and upgradation to intermediate lane of Paonta sahib Rajban Shillai Meenus Hatkoti road portion between km 97+000 to 106+120 (gumma to Fediz)(design rd 94+900 to 103+550) of NH 707 in the state of himachal pradesh”

- World bank documents “EMP for rehabilitation and upgradation to intermediate lane of Paonta sahib Rajban Shillai Meenus Hatkoti road portion between km 97+000 to 106+120 (Gumma to Fediz)(design road 94+900 to 103+550) of NH 707 in the state of Himachal Pradesh”

Thank You